

Sarthak Gupta

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EDUCATION

University of Florida

Bachelor of Science in Computer Science, Minor in Accounting (GPA: 3.68)

Gainesville, FL

Aug. 2024 – Dec. 2027

RELEVANT COURSEWORK

Introduction to Software Engineering, Data Structures and Algorithms, Operating Systems, Computer Organization, Information and Database Systems Design, Advanced Programming Fundamentals, Computational Linear Algebra, Discrete Structures, Introduction to Virtual Reality, Multivariable Calculus (III), Engineering Statistics, Business Finance

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Java, Python, JavaScript, C++, R, MATLAB

Frameworks & Platforms: TypeScript, Angular.js, React.js, Node.js, Express.js, LangChain, FastAPI, TensorFlow, ROS2, PyTorch, React Native, Expo, Firebase, MongoDB, GIS

Tools & Technologies: Git, NVIDIA Isaac Sim, OpenCV, OpenMV, AWS Polly, ComfyUI, Slurm, CUDA, Figma, Ubuntu/Linux, QGIS, ArcGIS Pro

Methodologies: SDLC, Agile, Scrum

Certifications: AWS Certified AI Practitioner, AWS Certified Cloud Practitioner, Building Transformer-Based NLP Applications (NVIDIA), Fundamentals of Deep Learning (NVIDIA), Intermediate Web Development (Code-Path)

EXPERIENCE

AI Software Engineering Intern

May 2026 – Aug. 2026

Qualcomm

San Diego, CA

- Worked within Qualcomm's **AI Software** organization on **on-device AI** for **Snapdragon** platforms, contributing to a pipeline spanning **voice**, **text**, and **image processing**.
- Gained exposure to how model execution is coordinated across on-device **voice**, **text**, and **image** workloads on Snapdragon hardware.
- Collaborated cross-functionally within a large-scale embedded AI software pipeline, working through onboarding, codebase ramp-up, and team development workflows at production scale.

Undergraduate Student Researcher

May 2025 – Present

Virtual Experiences Research Group (VERG) Lab @ University of Florida

Gainesville, FL

- Developing an end-to-end **AI video generation pipeline** in **ComfyUI** for the **U.S. Air Force**, producing cadet training content on sexual assault prevention.
- Deploying and scaling the pipeline on **HiPerGator HPC**, processing **500 GB** of raw video/image data and generating **200 training videos** with automated text, voice, and visual effects.
- Configuring **multi-node Slurm** jobs on HiPerGator, parallelizing **CUDA** workflows to accelerate AI model inference and video synthesis.
- Publication:** Murray, E.E., Canales, R., Venkatakrisnan, R., **Gupta, S.**, Lok, B., & Scaglione, N. (2026, February). *Leveraging generative AI's ability to adapt sexual assault prevention training for Air Force Cadets: Establishing proof of concept.* Orally presented internally at the annual **Digital Health Symposium**, University of Florida, Gainesville, FL.

AI Software Developer

Oct. 2025 – May 2026

Florida Community Innovation Foundation @ University of Florida

Gainesville, FL

- Designed and deployed **LLM-driven agentic pipelines** using **LangChain**, **OpenAI APIs**, and **Python** to autonomously extract, validate, and structure nonprofit and government service data from unstructured web sources.
- Built **web scraping** and **data-cleaning pipelines** processing over **5,000 resource listings** across food, housing, and mental health; improved data coverage by **35%**.

Robotics Software Engineer

Jan. 2025 – May 2026

Machine Intelligence Lab @ University of Florida

Gainesville, FL

- Developed a C++ Gazebo plugin within the **ROS2 control framework** for the SubjuGator 9 submarine's gripper, integrating **JointController** and **JointTrajectoryController** for **2-DOF** velocity and position control.
- Integrated a Water-Linked **Doppler Velocity Log (DVL)** using Bash, Linux, Python, and C++; raised localization update rate to **10 Hz** and cut drift from **12 m/hr** to **4 m/hr** in sea trials.
- Advanced SubjuGator 9 to the **semifinals of RoboSub 2025**, ranking **12th of 55** international teams via improved autonomy and sensor fusion.

COMPETITIONS

Vobile Innovation Challenge x IGNITE | Python, OpenCLAW, LangChain, Prompt Engineering

Apr. 2026

- Placed **1st** out of all competing teams at the **CADE Museum Students Thinking Big Event**.
- Engineered a multi-stage **prompt chaining pipeline** with **OpenCLAW** and **LangChain** to generate, evaluate, and refine creative concepts for Vobile's **DreamMaker** platform, enforcing consistency via **JSON mode** schemas and **few-shot exemplars**.
- Developed a **sliding-window visual grounding** algorithm generating video in **5-second segments**, re-injecting each clip's **reference keyframe** as a **visual context anchor**; eliminated cross-segment hallucination.

PROJECTS

JARVIS | Python, Gemini 2.5 Flash, openWakeWord, Whisper, Kokoro ONNX, PyQt6, Chrome MV3

2025 – Present

- Built a local, always-on voice assistant with a **multi-threaded daemon pipeline** (openWakeWord → Whisper STT → Gemini → Kokoro TTS), plus **Gmail push watch**, screen watching, and a **Chrome MV3** bridge streaming live DOM context.
- Designed a **two-tier memory** (short-term buffer + long-term JSON with top-K retrieval and importance decay) and a **vision loop** grounding screen actions on screenshots fused with DOM snapshots.